GreenForge AI

Sustainability Analysis Report

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Executive Summary

[Demo / Local AI mode] The sustainability impact shown is a modeled estimate based on benchmark runtime and configurable hardware assumptions (CPU TDP, PUE, grid carbon intensity). A reduction in runtime translates to less CPU-time consumed, which reduces estimated energy use. These are estimates — not validated environmental measurements. The primary benefit is reduced compute resource usage, which aligns with SDG 12 (Responsible Consumption).

GreenForge Score

Dimension	Score
Overall Score	72.2 / 100
Algorithmic Efficiency	60.0
Runtime Efficiency	60.0
Memory Efficiency	85.0
I/O Efficiency	90.0
ML Efficiency	80.0
Optimization Potential	76.0

Code Analysis Findings

Total: 5 issues detected — Critical: 0, High: 1, Medium: 4, Low: 0

[High] Nested Loop (depth 2) at line 4

A loop at line 4 contains an inner loop at line 5, creating a nesting depth of 2. If both loops iterate over input of size n, the inner body executes O(n²) or worse times. Whether this is a problem depends on expected input sizes and workload characteristics.

Recommendation: Investigate whether the inner loop can be replaced with: a hash lookup (O(1)), sorting + two-pointer (O(n log n)), set intersection, or vectorized operations (NumPy/Pandas). Only refactor if profiling confirms this is a bottleneck.

[Medium] Potentially repeated call to 'range()' inside loop at line 4

The function 'range()' is called inside a loop starting at line 4. If this call returns the same value on each iteration and has no side effects, computing it once before the loop may reduce unnecessary work.

Recommendation: If 'range()' is deterministic and the result does not change during the loop, hoist the call outside the loop and reuse the cached value.

[Medium] Potentially repeated call to 'len()' inside loop at line 4

The function 'len()' is called inside a loop starting at line 4. If this call returns the same value on each iteration and has no side effects, computing it once before the loop may reduce unnecessary work.

Recommendation: If 'len()' is deterministic and the result does not change during the loop, hoist the call outside the loop and reuse the cached value.

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Responsible AI Statement

All sustainability estimates in this report are modeled from benchmark measurements and configurable hardware/grid assumptions. They are indicative estimates, not validated environmental measurements. GreenForge AI does not claim exact carbon reduction. The GreenForge Score is a prototype indicator, not an industry certification.

SDG Alignment

Primary: SDG 12 — Responsible Consumption and Production
Secondary: SDG 13 — Climate Action
Secondary: SDG 9 — Industry, Innovation and Infrastructure